



RAG Pipeline

Enterprise v2.0 • Testleaf

DATA PIPELINE

Convert to JSON

Embeddings & Store

SEARCH ENGINES

Query Preprocessing

Vector Search

BM25 Search

Hybrid Search

Score Fusion

AI PROCESSING

Summarize & Dedup

Prompt & Schema

SYSTEM CONFIG

Settings

Enterprise AI Stack

MongoDB Atlas • Mistral AI • Groq

RAG Pipeline

Prompt & Schema

Configure system prompt templates

Mode: With RAG

Model: qwen/qwen3.8-72b

50 MongoDB Matches

3,843 / 8000 Tok (48% Budget Safe)

Evaluate Quality (DeepEval)

Export CSV

Generated Test Cases (3)

Retrieved MongoDB Context (8)

Linked User Stories (10)

Quality Validation & Approval Workflow

Manual Mode

Displays all 3 DeepEval metric scores. Review and manually click Approve (stores in MongoDB + creates in Jira) or Reject.

Manual Mode

Automation Mode

QUALITY THRESHOLDS:

Faithfulness ≥ 0.75

Contextual Recall ≥ 0.75

Hallucination ≤ 0.25 (target: 0)

Jira: Project KAN (Active Board)

Evaluated: 3/3

Validate All 3 Test Cases

Approve All Eligible (3)

Reject All

Manual Mode Active: Each test case row below displays all 3 metric scores. Review them and click Approve to save to MongoDB & create the Jira issue, or Reject to discard.

| ID      | Title                                                   | Module                              | Priority | Type        | Preconditions                                                                                                                                                                                                 | Test Steps                                                                                                                                                                                                                                                                                                                                                                                                                                        | Expected Results                                                                                                                                                                                                                                                                                                                                          | Quality Metrics                      |
|---------|---------------------------------------------------------|-------------------------------------|----------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| TC_6355 | Verify Secure WhatsApp Report Delivery After Validation | Patient Communication & Diagnostics | P1       | Integration | 1. Diagnostic technician user is logged in with appropriate permissions. 2. Patient record exists with a valid, consented WhatsApp number. 3. A lab report is in 'Pending Validation' status. 4. WhatsApp API | 1. Navigate to the Laboratory module and select the specific pending lab report for the patient. 2. Review the diagnostic results and click the 'Validate Report' button. 3. Confirm the validation action in the confirmation dialog. 4. Wait for the system to process the validation and trigger the communication service. 5. Verify the WhatsApp message received by the patient's device. 6. Click the secure download link provided in the | 1. Report status changes to 'Validated' immediately. 2. WhatsApp message is delivered within 30 seconds containing a unique secure link. 3. The link opens a secure viewer or downloads a PDF. 4. The PDF is encrypted and requires the temporary token provided in the link to open. 5. After 24 hours, the link returns a 'Link Expired' error message. | Faithfulness: Recall: Hallucination: |

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📁 All Bookmarks

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⚙️ Configure system prompt templates

⚙️

QUALITY THRESHOLDS: Faithfulness ≥ 0.75 Contextual Recall ≥ 0.75 Hallucination ≤ 0.25 (target: 0)

🔍 Jira: Project KAN (Active Board) Evaluated: 3/3

🔍 Validate All 3 Test Cases

✅ Approve All Eligible (3)

❌ Reject All

📄 Manual Mode Active: Each test case row below displays all 3 metric scores. Review them and click Approve to save to MongoDB & create the Jira issue, or Reject to discard.

| Priority | Type        | Preconditions                                                                                                                                                                              | Test Steps                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Expected Results                                                                                                                                                                                                                                                                                                                                                                                                        | Quality Metrics (DeepEval)                             | Gate Status | Approval & Jira |
|----------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|-------------|-----------------|
| P2       | Integration | 1. Patient record with valid consent and number.<br>2. WhatsApp API service is mocked to return a 503 Service Unavailable error.<br>3. Diagnostic report is in 'Pending Validation' state. | 8. Query the audit log for both events to verify specific error codes (e.g., CONSENT_MISSING, INVALID_PHONE).<br><br>1. Simulate WhatsApp API outage by configuring the test environment to return 503 errors.<br>2. Log in as a Diagnostic Technician.<br>3. Navigate to the pending lab report and click 'Validate Report'.<br>4. Observe the system response during the validation process.<br>5. Verify that the report status changes to 'Validated' despite the communication failure.<br>6. Check the system notification center or dashboard for a 'Delivery Failed' alert.<br>7. Restore the | 1. Report validation completes successfully; status updates to 'Validated'. 2. The communication service fails gracefully without crashing the main application. 3. A persistent alert is generated for the administrator or technician indicating 'WhatsApp Delivery Failed'. 4. Upon service restoration and retry, the message is successfully delivered to the patient. 5. All attempts (failed and successful) are | Faithfulness: N/A<br>Recall: N/A<br>Hallucination: N/A | Failed      | Approve Reject  |

The screenshot displays the RAG Pipeline Enterprise v2.0 - Testleaf application. The left sidebar contains navigation options: DATA PIPELINE (Convert to JSON, Embeddings & Store), SEARCH ENGINES (Query Preprocessing, Vector Search, BM25 Search, Hybrid Search, Score Fusion), AI PROCESSING (Summarize & Dedup), and SYSTEM CONFIG (Settings). The main area shows the 'Prompt & Schema' configuration page.

### Quality Validation & Approval Workflow

**Automation Mode**

Auto-validates generated test cases against DeepEval thresholds; publishes to MongoDB & Jira only when all 3 pass.

**QUALITY THRESHOLDS:**

- Faithfulness ≥ 0.75
- Contextual Recall ≥ 0.75
- Hallucination ≤ 0.25 (target: 0)

Jira: Project KAN (Active Board) | Evaluated: 3/3

+ Validate All 3 Test Cases

**AUTOMATION PUBLISHING BLOCKED: QUALITY THRESHOLD FAILED**

One or more test cases did not satisfy all 3 DeepEval quality metrics (Faithfulness ≥ 0.75, Contextual Recall ≥ 0.75, Hallucination ≤ 0.25). Publishing to MongoDB and Jira has been stopped.

- [TC.6355] Verify Secure WhatsApp Report Delivery After Validation:
  - Faithfulness: N/A < 0.75 threshold (Factually inconsistent with context)
  - Contextual Recall: N/A < 0.75 threshold (Missing required context coverage)
  - Hallucination: N/A > 0.25 threshold (Contradicts or ungrounded in context)
- [TC.6356] Verify Consent Verification and Invalid Number Handling:
  - Faithfulness: N/A < 0.75 threshold (Factually inconsistent with context)
  - Contextual Recall: N/A < 0.75 threshold (Missing required context coverage)
  - Hallucination: N/A > 0.25 threshold (Contradicts or ungrounded in context)
- [TC.6357] Verify System Resilience During WhatsApp Service Outage:
  - Faithfulness: N/A < 0.75 threshold (Factually inconsistent with context)
  - Contextual Recall: N/A < 0.75 threshold (Missing required context coverage)
  - Hallucination: N/A > 0.25 threshold (Contradicts or ungrounded in context)

Publishing stopped automatically. No documents were stored in MongoDB and no Jira tickets were created.

| Priority | Type | Preconditions | Test Steps                                                                                                                                                                                                                             | Expected Results                                                                                                                                                            | Quality Metrics (DeepEval) | Gate Status | Automation Gate / Jira |
|----------|------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-------------|------------------------|
|          |      |               | Configuring the test environment to return 503 errors.<br>2. Log in as a Diagnostic Technician.<br>3. Navigate to the pending lab report and click 'Validate Report'.<br>4. Observe the system response during the validation process. | 1. Report validation completes successfully; status updates to 'Validated'.<br>2. The communication service fails gracefully without crashing the main application.<br>3. A |                            |             |                        |

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All Bookmarks

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MongoDB Atlas • Mistral AI • Groq

⏪ RAG Pipeline > 📄 Prompt & Schema

⚙️ Configure system prompt templates

QUALITY THRESHOLDS: 

Faithfulness ≥ 0.75

Contextual Recall ≥ 0.75

Hallucination ≤ 0.25 (target: 0)

🔍 Validate All 2 Test Cases

👍 Approve All Eligible (1)

🚫 Reject All

📄 1 Published to MongoDB & Jira

🕒 Manual Mode Active: Each test case row below displays all 3 metric scores. Review them and click Approve to save to MongoDB & create the Jira issue, or Reject to discard.

| Module     | Priority | Type       | Preconditions                                                                                                                                                                                                       | Test Steps                                                                                                                                                                                                                                                                                                                                                                                                                                        | Expected Results                                                                                                                                                                 | Quality Metrics (DeepEval)                             | Gate Status | Approval & Jira                                       |
|------------|----------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|-------------|-------------------------------------------------------|
| Laboratory | P1       | Functional | 1. An alert has been entered and flagged. 2. The alert has been broadcast to the attending physician. 3. Pathologist login credentials are available. 4. Audit log access permissions are granted to the QA tester. | 5. Verify that the system prevents sign-off and displays an error message: 'Digital Signature Mandatory for Critical Values'. 6. Enter the required digital signature (PIN/Password) and click 'Sign Off'. 7. Verify that the report status changes to 'Signed' and the sign-off timestamp is recorded. 8. Navigate to the Audit Log module and verify an entry exists for the alert acknowledgement and sign-off event with the pathologist's ID | is displayed clearly. 3. Upon valid signature, the report is marked as 'Signed'. 4. Audit log contains a record of the sign-off action with user ID, timestamp, and action type. | Faithfulness: N/A<br>Recall: N/A<br>Hallucination: N/A | 🟢 Approved  | <div>🔗 KAN-6</div> <div>✓ In MongoDB &amp; Jira</div> |



